

National Research University “Higher School of Economics”  
Faculty of Computer Science

Bachelor’s Programme “Data Science and Business Analytics”  
Field of Study: 01.03.02 Applied Mathematics and Information Science

## **BACHELOR’S THESIS**

### **Bitcoin Price Prediction Using Machine Learning Models and Stochastic Differential Equations**

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Moscow, 2026

# Abstract

This thesis investigates Bitcoin price prediction through two complementary lenses: classical stochastic differential equation (SDE) models from mathematical finance, and modern machine learning (ML) methods. A key theoretical contribution is the rigorous justification of the stochastic process representation of cryptocurrency prices, grounded in the Efficient Market Hypothesis, the Random Walk Hypothesis, and the empirical stylised facts of Bitcoin log-returns (heavy tails, volatility clustering, jump events).

Three SDE models are derived, calibrated, and evaluated: Geometric Brownian Motion (GBM), the Merton Jump-Diffusion model, and the Heston Stochastic Volatility model. Four ML models — Elastic Net, Random Forest, Gradient Boosting, and Long Short-Term Memory (LSTM) networks — are trained and evaluated on multi-resolution Bitcoin price data obtained from Yahoo Finance.

As the primary *novel contribution*, a **Hybrid SDE-ML model** is proposed and implemented. The hybrid model replaces the constant drift of GBM with a time-varying drift predicted at each step by a Gradient Boosting regressor, while retaining the stochastic diffusion term scaled by rolling realised volatility. This architecture combines the distributional forecasting capability of SDEs with the predictive accuracy of ML.

All models are evaluated under **walk-forward cross-validation** (five folds, expanding window), which prevents data leakage and provides robust out-of-sample estimates. Beyond standard regression metrics (MAE, RMSE,  $R^2$ ), an **economic evaluation** is conducted via a simple momentum-driven trading strategy, reporting annualised Sharpe ratios and maximum drawdown.

The empirical results show that feature-based ML models are very strong short-horizon baselines, while the Hybrid SDE-ML framework remains competitive and adds calibrated 90 % prediction intervals (coverage 91.7 %) together with an economically interpretable stochastic structure.

**Keywords:** Bitcoin, cryptocurrency, stochastic differential equations, Geometric Brownian Motion, Merton model, Heston model, machine learning, LSTM, hybrid model, time-varying drift, walk-forward validation, Sharpe ratio.

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# Introduction

## Motivation

Bitcoin was introduced in 2008 by the pseudonymous Satoshi Nakamoto as the first decentralised digital currency [1]. In the sixteen years since its creation, Bitcoin has grown from an obscure cryptographic experiment into a global asset class with a market capitalisation exceeding one trillion US dollars. Its price exhibits a unique combination of extreme volatility, structural breaks, speculative bubbles, and rapid recoveries that challenge traditional financial modelling paradigms.

The question of whether Bitcoin prices can be predicted — and, if so, by what class of models — has attracted considerable academic interest. From the perspective of mathematical finance, the natural starting point is to ask: *Is Bitcoin price a stochastic process, and if so, which one?* Answering this question rigorously requires both theoretical arguments and empirical evidence derived from the statistical properties of observed price data.

## Problem Statement

Let  $S_t$  denote the Bitcoin price in US dollars at time  $t$ . The fundamental modelling problem may be stated as follows.

1. **(Theoretical)** Under which assumptions on market structure and investor behaviour does  $\{S_t\}_{t \geq 0}$  admit a representation as the solution of a stochastic differential equation?
2. **(Statistical)** Do the empirical log-returns of Bitcoin,  $r_t = \ln(S_t/S_{t-1})$ , satisfy the probabilistic conditions implied by candidate SDE models?
3. **(Predictive)** Among SDE models, ML models, and their hybrids, which achieves the best out-of-sample forecasting performance under rigorous walk-forward validation?
4. **(Economic)** Do the predictive gains of ML and hybrid models translate into economically meaningful improvements in trading performance, as measured by the Sharpe ratio?

## Research Objectives

- to provide a self-contained introduction to stochastic processes, Itô calculus, and SDEs;

- to rigorously justify the stochastic process representation of Bitcoin prices via propositions and proofs;
- to derive, calibrate, and evaluate three SDE models: GBM, Merton Jump-Diffusion, and Heston;
- to build, train, and evaluate four ML models including LSTM;
- to propose and implement a novel **Hybrid SDE-ML model** with time-varying ML-predicted drift;
- to conduct rigorous walk-forward cross-validation;
- to evaluate all models on an economic criterion (Sharpe ratio).

## Structure of the Thesis

Chapter 1 develops the mathematical foundations. Chapter 2 justifies the stochastic process framework for Bitcoin. Chapter 3 derives and analyses the three SDE models. Chapter 4 describes the ML models and the novel Hybrid SDE-ML model. Chapter 5 covers data, feature engineering, walk-forward validation, and evaluation protocols including economic metrics. Chapter 6 presents the experimental results. The Conclusion summarises findings and outlines future work.

# Chapter 1

## Mathematical Foundations of Stochastic Processes

### 1.1 Probability Spaces and Filtrations

**Definition 1.1** (Probability Space). A *probability space* is a triple  $(\Omega, \mathcal{F}, \mathbb{P})$  where  $\Omega$  is a non-empty set of elementary outcomes,  $\mathcal{F}$  is a  $\sigma$ -algebra of subsets of  $\Omega$  (events), and  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$  is a probability measure satisfying  $\mathbb{P}(\Omega) = 1$  and countable additivity.

Financial modelling requires tracking the evolution of information over time.

**Definition 1.2** (Filtration). Let  $T > 0$ . A *filtration* on  $(\Omega, \mathcal{F}, \mathbb{P})$  is a family  $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$  of sub- $\sigma$ -algebras of  $\mathcal{F}$  satisfying  $\mathcal{F}_s \subseteq \mathcal{F}_t$  for all  $s \leq t$ . The filtered probability space  $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$  is said to satisfy the *usual conditions* if  $\mathcal{F}_0$  contains all  $\mathbb{P}$ -null sets and the filtration is right-continuous:  $\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s$  for all  $t$ .

**Definition 1.3** (Stochastic Process). A *stochastic process* on  $(\Omega, \mathcal{F}, \mathbb{P})$  is a collection of random variables  $\{X_t\}_{t \in \mathcal{T}}$  indexed by a parameter set  $\mathcal{T}$  (typically  $[0, T]$  or  $\mathbb{Z}_{\geq 0}$ ). The process is said to be  $\mathbb{F}$ -*adapted* if  $X_t$  is  $\mathcal{F}_t$ -measurable for every  $t$ .

The concept of adaptedness captures the intuition that at time  $t$ , the value  $X_t$  is determined by the information available up to that moment.

### 1.2 Martingales

**Definition 1.4** (Martingale). An adapted, integrable process  $\{M_t\}$  is a *martingale* with respect to  $\mathbb{F}$  if

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{for all } 0 \leq s \leq t.$$

It is a *supermartingale* (resp. *submartingale*) if the equality is replaced by  $\leq$  (resp.  $\geq$ ).

The martingale property is intimately related to the notion of a fair game: in a frictionless arbitrage-free market, discounted asset prices are martingales under a risk-neutral probability measure.



This observation, formalised in the First Fundamental Theorem of Asset Pricing, provides one of the deepest theoretical justifications for modelling asset prices as stochastic processes.

### 1.3 The Wiener Process (Brownian Motion)

The Wiener process is the canonical building block of continuous-time stochastic models in finance.

**Definition 1.5** (Standard Wiener Process). A stochastic process  $\{W_t\}_{t \geq 0}$  on a filtered probability space  $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$  is a *standard Wiener process* (or *standard Brownian motion*) if:

(W1)  $W_0 = 0$  almost surely;

(W2)  $W$  has *independent increments*: for  $0 \leq t_0 < t_1 < \dots < t_n$ , the increments  $W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$  are mutually independent;

(W3)  $W$  has *stationary Gaussian increments*: for all  $0 \leq s < t$ ,

$$W_t - W_s \sim \mathcal{N}(0, t - s);$$

(W4)  $W$  has *continuous paths*:  $t \mapsto W_t(\omega)$  is continuous for  $\mathbb{P}$ -almost every  $\omega \in \Omega$ .

**Proposition 1.6** (Martingale Property of Brownian Motion). A standard Wiener process  $\{W_t\}$  is a martingale with respect to its natural filtration  $\mathcal{F}_t^W = \sigma(W_s : 0 \leq s \leq t)$ .

*Proof.* For  $s \leq t$ , write  $W_t = W_s + (W_t - W_s)$ . By (W2),  $(W_t - W_s)$  is independent of  $\mathcal{F}_s^W$ . By (W3),  $\mathbb{E}[W_t - W_s] = 0$ . Therefore,

$$\mathbb{E}[W_t \mid \mathcal{F}_s^W] = W_s + \mathbb{E}[W_t - W_s \mid \mathcal{F}_s^W] = W_s + \mathbb{E}[W_t - W_s] = W_s. \quad \square$$

**Proposition 1.7** (Quadratic Variation of Brownian Motion). For a standard Wiener process, the quadratic variation over  $[0, t]$  satisfies

$$\langle W \rangle_t := \lim_{\|\Pi\| \rightarrow 0} \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 = t \quad \text{in } L^2(\mathbb{P}),$$

where  $\Pi = \{0 = t_0 < t_1 < \dots < t_n = t\}$  and  $\|\Pi\| = \max_i (t_i - t_{i-1})$ .

This non-trivial result — that Brownian paths accumulate quadratic variation at the constant rate of one unit per unit of time — is the foundation of Itô's calculus. Informally, the notation  $dW_t^2 = dt$  encodes this fact.

*Remark 1.8.* Brownian motion paths are continuous but nowhere differentiable with probability one. Consequently, the classical Lebesgue-Stieltjes integral  $\int_0^t f(s) dW_s$  is not well-defined pathwise, and a new integration theory — the Itô integral — is required.

## 1.4 The Itô Stochastic Integral

**Definition 1.9** (Itô Integral for Simple Processes). Let  $\phi = \sum_{i=0}^{n-1} \phi_i \mathbb{1}_{[t_i, t_{i+1})}$  be an  $\mathbb{F}$ -adapted step process with  $\mathbb{E}[\phi_i^2] < \infty$ . The *Itô integral* of  $\phi$  with respect to  $W$  is defined as

$$I_t(\phi) := \int_0^t \phi_s dW_s = \sum_{i: t_i < t} \phi_i (W_{t_{i+1} \wedge t} - W_{t_i}).$$

For general adapted square-integrable processes  $\phi$ , the integral is extended by a density argument in the  $L^2$  sense. The key isometry property is stated below.

**Proposition 1.10** (Itô Isometry). *For any  $\mathbb{F}$ -adapted, square-integrable process  $\phi$ ,*

$$\mathbb{E} \left[ \left( \int_0^t \phi_s dW_s \right)^2 \right] = \int_0^t \mathbb{E}[\phi_s^2] ds.$$

This identity implies that  $\{I_t(\phi)\}_{t \geq 0}$  is a martingale, and underpins the  $L^2$  theory of stochastic integration.

## 1.5 Itô's Lemma

Itô's lemma is the stochastic analogue of the chain rule and is the single most important computational tool in continuous-time finance.

**Theorem 1.11** (Itô's Lemma — One-Dimensional Form). *Let  $X_t$  satisfy the stochastic differential equation  $dX_t = \mu_t dt + \sigma_t dW_t$ , and let  $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$  be a function of class  $C^{1,2}$  (once continuously differentiable in  $t$ , twice in  $x$ ). Then*

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \sigma_t^2 dt. \quad (1.1)$$

The extra term  $\frac{1}{2}(\partial^2 f / \partial x^2) \sigma_t^2 dt$  arises precisely because of Proposition 1.7: when we expand  $df$  using Taylor's theorem, the term  $(\partial^2 f / \partial x^2)(dX_t)^2 / 2$  contributes  $\frac{1}{2}(\partial^2 f / \partial x^2) \sigma_t^2 dt$  instead of vanishing as it would in classical calculus.

## 1.6 Existence and Uniqueness for SDEs

**Definition 1.12** (Stochastic Differential Equation). A *stochastic differential equation* (SDE) on  $[0, T]$  is of the form

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x_0 \in \mathbb{R}, \quad (1.2)$$

where  $b, \sigma : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$  are measurable coefficient functions.

**Theorem 1.13** (Existence and Uniqueness — Lipschitz Coefficients). *Suppose the coefficients  $b$  and  $\sigma$  in (1.2) satisfy:*

(C1) **Global Lipschitz condition:** *there exists  $L > 0$  such that*

$$|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq L|x - y|$$

*for all  $t \in [0, T]$  and  $x, y \in \mathbb{R}$ ;*

(C2) **Linear growth condition:** *there exists  $C > 0$  such that*

$$|b(t, x)|^2 + |\sigma(t, x)|^2 \leq C^2(1 + x^2)$$

*for all  $t \in [0, T]$  and  $x \in \mathbb{R}$ .*

*Then equation (1.2) has a unique strong solution  $\{X_t\}_{t \in [0, T]}$  that is  $\mathbb{F}$ -adapted and satisfies  $\mathbb{E}[\sup_{0 \leq t \leq T} X_t^2] < \infty$ .*

We will verify conditions (C1)–(C2) for each of the three SDE models studied in Chapter 3.

# Chapter 2

## Why Bitcoin Price Is a Stochastic Process

This chapter presents the theoretical and empirical justifications for treating the Bitcoin price  $S_t$  as the solution of a stochastic differential equation. The argument proceeds in three stages.

1. **Market microstructure.** Under the Efficient Market Hypothesis, asset prices incorporate all available information instantaneously, implying that price changes are driven by new, unpredictable information arrivals — naturally modelled as random shocks.
2. **Random Walk Hypothesis.** The discrete-time log-return series  $\{r_t\}$  has the statistical properties of an i.i.d. process, at least to a first approximation.
3. **Stylised facts.** The empirical distribution of Bitcoin log-returns exhibits heavy tails, volatility clustering, and occasional large jumps — properties that motivate extensions of the basic Gaussian random walk.

### 2.1 The Efficient Market Hypothesis

**Definition 2.1** (Informational Efficiency [2]). A financial market is said to be *efficient* with respect to an information set  $\Phi_t$  if asset prices fully and instantaneously reflect all information contained in  $\Phi_t$ .

Three nested forms are commonly distinguished.

- **Weak-form efficiency:**  $\Phi_t$  contains only past prices and trading volumes.
- **Semi-strong efficiency:**  $\Phi_t$  includes all publicly available information (financial statements, macroeconomic data, news).
- **Strong-form efficiency:**  $\Phi_t$  includes all information, public and private.

**Proposition 2.2** (Stochastic Prices under the EMH). *Suppose the market for Bitcoin satisfies weak-form efficiency. Then the price process  $\{S_t\}$  is a Markov process, and the log-return process  $\{r_t = \ln(S_t/S_{t-1})\}$  constitutes a sequence of random variables uncorrelated with any function of past prices.*

*Argument.* Under weak-form efficiency, no trading strategy based solely on the history  $(S_0, S_1, \dots, S_{t-1})$  can generate expected profits above the equilibrium return. Formally, if  $\pi_t$  denotes the equilibrium expected return per period,

$$\mathbb{E}[S_{t+1} \mid S_0, \dots, S_t] = (1 + \pi_t)S_t.$$

This is the martingale (or supermartingale) property. Since  $S_{t+1}$  depends on past prices only through  $S_t$ , the process is Markov. Any excess return  $r_t - \pi_{t-1}$  is unpredictable from past information, hence random.  $\square$

*Remark 2.3.* Proposition 2.2 does *not* claim that prices are unpredictable in an absolute sense — only that the predictable component is the risk premium. The unpredictable residual, the “news component”, is what stochastic models aim to characterise.

## 2.2 The Random Walk Hypothesis

In continuous time, the “price driven by random news arrivals” idea leads to the following hypothesis.

**Assumption 2.4** (Continuous Random Walk Hypothesis). The log-price process  $\{L_t = \ln S_t\}$  satisfies

$$L_t - L_s = \mu(t - s) + \sigma(W_t - W_s), \quad 0 \leq s \leq t,$$

for constants  $\mu \in \mathbb{R}$  (drift) and  $\sigma > 0$  (diffusion coefficient), where  $W$  is a standard Wiener process.

**Proposition 2.5** (Equivalence with Geometric Brownian Motion). *Assumption 2.4 is equivalent to the requirement that  $S_t$  solves the SDE*

$$dS_t = \mu^* S_t dt + \sigma S_t dW_t, \quad \mu^* = \mu + \frac{1}{2}\sigma^2,$$

*i.e.,  $S_t$  follows Geometric Brownian Motion.*

*Proof.* Apply Itô’s Lemma (Theorem 1.11) to  $f(x) = e^x$  with  $X_t = \ln S_t = L_t$ . From Assumption 2.4,  $dL_t = \mu dt + \sigma dW_t$ . Then

$$dS_t = d(e^{L_t}) = e^{L_t} dL_t + \frac{1}{2}e^{L_t}(dL_t)^2 = S_t(\mu dt + \sigma dW_t) + \frac{1}{2}S_t\sigma^2 dt = (\mu + \frac{1}{2}\sigma^2)S_t dt + \sigma S_t dW_t. \quad \square$$

### 2.2.1 Empirical Testing of the Random Walk Hypothesis

The Random Walk Hypothesis (RWH) implies specific testable restrictions on the log-return series. Three classical tests are considered here.

**Augmented Dickey-Fuller test.** The ADF test examines whether the log-price series contains a unit root. Under the null  $H_0: \rho = 1$ , the log-price follows a random walk (possibly with drift). For Bitcoin, empirical studies consistently *fail to reject*  $H_0$  at standard significance levels for prices ( $p = 0.43$ ), while the return series  $\{r_t\}$  is stationary ( $p < 0.001$ ).

**Ljung-Box test for serial correlation.** The Ljung-Box  $Q$ -statistic tests the joint hypothesis that the first  $k$  autocorrelations of  $\{r_t\}$  are zero:

$$Q(k) = n(n+2) \sum_{j=1}^k \frac{\hat{\rho}_j^2}{n-j},$$

where  $\hat{\rho}_j$  is the sample autocorrelation at lag  $j$  and  $n$  is the sample size. Under the RWH,  $Q(k) \xrightarrow{d} \chi^2(k)$ . Empirical results for Bitcoin show negligible autocorrelation in raw returns, consistent with the RWH, but *significant* autocorrelation in squared returns  $\{r_t^2\}$  ( $p < 0.001$  at lags 1–10) — the signature of volatility clustering discussed in Section 2.3.

**Variance ratio test.** If  $\{r_t\}$  is i.i.d., the variance of  $k$ -period returns equals  $k$  times the variance of one-period returns. The variance ratio

$$VR(k) = \frac{\text{Var}[r_t^{(k)}]}{k \cdot \text{Var}[r_t]}, \quad r_t^{(k)} = \sum_{j=0}^{k-1} r_{t-j},$$

should equal one under the RWH. Empirical estimates for daily Bitcoin data typically yield  $VR(k) \approx 1$  for moderate  $k$ , supporting the RWH in the first-moment sense, although deviations appear at high frequencies.

## 2.3 Stylised Facts of Bitcoin Log-Returns

While the RWH provides a theoretical justification for stochastic modelling, the Gaussian random walk (Assumption 2.4) is an idealisation that fails to capture several well-documented empirical regularities, known as *stylised facts*.

### 2.3.1 Heavy Tails (Leptokurtosis)

**Proposition 2.6** (Kurtosis of Gaussian Returns). *If  $\{r_t\} \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$ , then the excess kurtosis is  $\kappa = 0$ .*

In practice, the excess kurtosis of Bitcoin daily log-returns is  $\hat{\kappa} \approx 7.84$ , implying a probability of large moves far in excess of Gaussian predictions. A Jarque-Bera test strongly rejects normality at any conventional significance level ( $p < 0.001$ ).

The heavy-tailed distribution of returns can be attributed to two sources.

- **Jump events:** rare but large discontinuous price movements caused by exchange hacks, regulatory announcements, or macroeconomic shocks. The Merton Jump-Diffusion model (Section 3.2) captures this by superimposing a compound Poisson process on the Brownian diffusion.
- **Stochastic volatility:** if the instantaneous variance  $v_t$  is itself random and time-varying, the marginal distribution of returns  $r_t = \int_{t-\Delta}^t \sqrt{v_s} dW_s + \mu\Delta$  is a Gaussian scale mixture, which is inherently heavy-tailed. This motivates the Heston model (Section 3.3).

### 2.3.2 Volatility Clustering

**Definition 2.7** (Volatility Clustering). A return series  $\{r_t\}$  exhibits *volatility clustering* if large (small) absolute returns tend to be followed by large (small) absolute returns:

$$\text{Corr}(|r_t|, |r_{t-k}|) > 0 \quad \text{for small } k \geq 1.$$

Volatility clustering was first documented by Mandelbrot [3] for commodity prices and has been universally confirmed in cryptocurrency markets. It implies that the instantaneous volatility  $\sigma_t$  is not constant but is itself a stochastic process, providing direct empirical motivation for the Heston model.

**Proposition 2.8** (GBM Cannot Exhibit Volatility Clustering). *Under GBM, the log-returns  $r_t = (\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\Delta W$  are i.i.d. Gaussian, so  $\text{Corr}(|r_t|, |r_{t-k}|) = 0$  for all  $k \geq 1$ .*

This proposition shows that GBM is *structurally incapable* of reproducing volatility clustering, and motivates the introduction of stochastic volatility models.

### 2.3.3 Leverage Effect (Asymmetric Volatility)

Empirical studies of traditional equity markets report a *negative* correlation between return shocks and subsequent volatility changes: large negative returns tend to be followed by elevated volatility. For Bitcoin, this effect is attenuated but still statistically significant, particularly at daily frequencies.

Formally, if  $\rho = \text{Corr}(dW_1, dW_2)$  denotes the correlation between the price Brownian  $W_1$  and the variance Brownian  $W_2$  in the Heston model, then  $\rho < 0$  captures the leverage effect.

### 2.3.4 Non-Stationarity and Structural Breaks

Bitcoin prices have undergone several distinct regimes, including explosive growth phases (2017, 2020-2021) and sustained drawdowns (2018, 2022). The log-price series is non-stationary ( $I(1)$ ), while the log-return series is weakly stationary. Structural break tests (e.g., the Bai-Perron test) identify multiple change-points in the drift parameter  $\mu$  and the volatility parameter  $\sigma$ .

*Remark 2.9.* Non-stationarity of the level process  $\{S_t\}$  is *consistent* with the GBM model, since GBM is a continuous-time process with non-stationary levels and (weakly) stationary log-returns. However, regime changes in  $\mu$  and  $\sigma$  suggest that a constant-coefficient SDE is a simplification, and that time-varying or regime-switching models may be more appropriate for long horizons. This observation directly motivates the Hybrid SDE-ML model introduced in Section 4.6.

## 2.4 Why the Stochastic Process Framework is Appropriate

We now synthesise the foregoing arguments into a formal justification.

**Proposition 2.10** (Stochastic Process Representation). *Suppose the following conditions hold for the Bitcoin market.*

- (A1) (**Weak-form efficiency**) No strategy based on price history generates risk-free profits above the equilibrium return.
- (A2) (**Information continuity**) New information arrives continuously and in small doses rather than in a finite number of large discrete jumps.
- (A3) (**Finite variance**) The log-return process has finite second moments over any fixed time horizon.
- (A4) (**Independent information shocks**) The incremental information arriving in non-overlapping time intervals is independent.

Then there exists a probability space  $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$  and a standard Wiener process  $W$  on it, such that the log-price process  $\{L_t = \ln S_t\}$  can be represented as

$$L_t = L_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s,$$

for  $\mathbb{F}$ -adapted processes  $\mu$  and  $\sigma > 0$  with  $\int_0^T (\mu_s^2 + \sigma_s^2) ds < \infty$  a.s.

*Sketch of proof.* Assumption (A1) implies that discounted prices are martingales under the physical measure (up to a risk premium). By the Martingale Representation Theorem (see [4]), any square-integrable martingale on a Brownian filtration admits the stochastic integral representation above. Assumption (A2) ensures the filtration is generated by a Brownian motion rather than a Poisson or jump process. Assumptions (A3)–(A4) ensure the coefficients are well-defined and the integral converges.  $\square$

*Remark 2.11.* The departure from Assumption (A2) — allowing large discrete jumps — leads naturally to the Merton Jump-Diffusion model (Section 3.2), where the driving noise is  $W_t + \sum_{i=1}^{N_t} J_i$ , a superposition of a Brownian motion and a compound Poisson process.



# Chapter 3

## Stochastic Differential Equation Models for Bitcoin

### 3.1 Geometric Brownian Motion

#### 3.1.1 Model Formulation

**Assumption 3.1** (GBM Assumptions). (G1) The market is frictionless (no transaction costs, taxes, or short-selling restrictions).

(G2) The log-return over any infinitesimal interval  $dt$  is independently and identically Gaussian-distributed:  $d \ln S_t \sim \mathcal{N}(v dt, \sigma^2 dt)$  for constants  $v \in \mathbb{R}$ ,  $\sigma > 0$ .

(G3) The drift  $v$  and volatility  $\sigma$  are constant over the modelling horizon.

**Proposition 3.2** (GBM SDE). *Under Assumption 3.1, the price process satisfies*

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (3.1)$$

where  $\mu = v + \frac{1}{2}\sigma^2$  is the expected instantaneous rate of return.

*Proof.* The hypothesis  $d \ln S_t = v dt + \sigma dW_t$  means  $\ln S_t = \ln S_0 + vt + \sigma W_t$ . Setting  $f(x) = e^x$  and applying Itô's Lemma (Theorem 1.11):

$$dS_t = S_t d(\ln S_t) + \frac{1}{2} S_t (d \ln S_t)^2 = S_t (v dt + \sigma dW_t) + \frac{1}{2} S_t \sigma^2 dt = (v + \frac{1}{2}\sigma^2) S_t dt + \sigma S_t dW_t.$$

Setting  $\mu = v + \frac{1}{2}\sigma^2$  gives equation (3.1). □

#### 3.1.2 Closed-Form Solution

**Proposition 3.3** (Explicit Solution to GBM). *The unique strong solution to (3.1) is*

$$S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}. \quad (3.2)$$

In particular,  $\ln(S_t/S_0) \sim \mathcal{N}((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t)$ .

*Proof.* The coefficients  $b(t, x) = \mu x$  and  $\sigma(t, x) = \sigma x$  satisfy the global Lipschitz and linear growth conditions of Theorem 1.13 after a logarithmic change of variables. The closed form follows from integrating  $d \ln S_t = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW_t$ .  $\square$

### 3.1.3 Parameter Calibration

Given a discrete price series  $S_0, S_1, \dots, S_n$  sampled at intervals  $\Delta t$ , the log-returns  $r_i = \ln(S_i/S_{i-1}) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2 \Delta t)$ . The maximum likelihood estimators are

$$\hat{\sigma}^2 = \frac{1}{(n-1)\Delta t} \sum_{i=1}^n (r_i - \bar{r})^2, \quad \hat{\mu} = \frac{\bar{r}}{\Delta t} + \frac{\hat{\sigma}^2}{2}, \quad (3.3)$$

where  $\bar{r} = n^{-1} \sum_{i=1}^n r_i$ . Both estimators are consistent and asymptotically normal under the model.

### 3.1.4 Limitations for Bitcoin

GBM captures only the first and second moments of the return distribution. For Bitcoin, the following empirical features are inconsistent with GBM.

- Excess kurtosis ( $\hat{\kappa} > 0$ ) implies fatter tails than Gaussian.
- Significant autocorrelation of squared returns implies non-constant volatility.
- Occasional “crash” or “bubble” events imply the presence of large discrete jumps.

These limitations motivate the more sophisticated models developed in Sections 3.2 and 3.3.

## 3.2 Merton Jump-Diffusion Model

### 3.2.1 Motivation and Assumptions

The Merton (1976) model [5] augments GBM by a compound Poisson process that accounts for sudden, large price movements caused by unexpected news events. Such events are particularly common in the cryptocurrency market: exchange hacks, regulatory bans, and large institutional trades can cause single-day price movements of 10 %–30 % that are impossible under GBM.

**Assumption 3.4** (Merton Jump-Diffusion Assumptions). (M1) Between jumps, the price process follows GBM with constant drift  $\mu$  and diffusion coefficient  $\sigma$ .

(M2) Jumps arrive according to a homogeneous Poisson process  $\{N_t\}$  with constant intensity  $\lambda > 0$ , independent of  $W_t$ .

(M3) The  $k$ -th jump multiplies the price by a random factor  $Y_k$ , where  $\ln Y_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_J, \sigma_J^2)$ , independent of each other and of  $N_t$  and  $W_t$ .

(M4) The expected relative jump size is  $\kappa = \mathbb{E}[Y_k - 1] = e^{\mu_J + \sigma_J^2/2} - 1$ .

### 3.2.2 Model Formulation

**Proposition 3.5** (Merton SDE). *Under Assumption 3.4, the price process satisfies*

$$dS_t = (\mu - \lambda \kappa) S_{t-} dt + \sigma S_{t-} dW_t + S_{t-} dJ_t, \quad (3.4)$$

where  $J_t = \sum_{k=1}^{N_t} (Y_k - 1)$  is the compound Poisson process,  $S_{t-} = \lim_{s \nearrow t} S_s$  is the left-limit, and the drift is adjusted by  $-\lambda \kappa S_{t-} dt$  to ensure  $\mathbb{E}[dS_t] = \mu S_{t-} dt$ .

*Proof.* The total price change over  $[t, t + dt]$  consists of a continuous diffusion part and a jump part. The expected contribution of the compound Poisson process over  $dt$  is  $S_{t-} \mathbb{E}[dJ_t] = S_{t-} \lambda \kappa dt$ . To ensure the total expected return equals  $\mu dt$ , the continuous drift is set to  $(\mu - \lambda \kappa) S_{t-} dt$ .  $\square$

### 3.2.3 Return Distribution

**Proposition 3.6** (Return Distribution under the Merton Model). *Under the Merton Jump-Diffusion model, the log-return over horizon  $\Delta t$  is*

$$\ln \frac{S_{t+\Delta t}}{S_t} = \left( \mu - \lambda \kappa - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z + \sum_{k=1}^{N(\Delta t)} \ln Y_k, \quad (3.5)$$

where  $Z \sim \mathcal{N}(0, 1)$  and  $N(\Delta t) \sim \text{Poisson}(\lambda \Delta t)$ , all independent. Conditionally on  $N(\Delta t) = n$ , the log-return is Gaussian:

$$\ln(S_{t+\Delta t}/S_t) \mid N(\Delta t) = n \sim \mathcal{N}(m_n \Delta t, v_n \Delta t), \quad (3.6)$$

where  $m_n = \mu - \lambda \kappa - \frac{1}{2} \sigma^2 + n \mu_J / \Delta t$  and  $v_n = \sigma^2 + n \sigma_J^2 / \Delta t$ .

The unconditional distribution is therefore a *Poisson mixture of Gaussians*, which has heavier tails than a single Gaussian for any  $\lambda > 0$ ,  $\sigma_J > 0$ . This directly addresses the leptokurtosis observed in Bitcoin returns.

### 3.2.4 Maximum Likelihood Calibration

The log-likelihood function for the observed return vector  $\mathbf{r} = (r_1, \dots, r_n)$ ,  $r_i = \ln(S_i/S_{i-1})$ , is

$$\ell(\theta; \mathbf{r}) = \sum_{i=1}^n \ln \left[ \sum_{k=0}^{\infty} \frac{e^{-\lambda \Delta t} (\lambda \Delta t)^k}{k!} \cdot \phi(r_i; m_k \Delta t, v_k \Delta t) \right], \quad (3.7)$$

where  $\phi(\cdot; \bar{m}, \bar{v})$  denotes the Gaussian density with mean  $\bar{m}$  and variance  $\bar{v}$ , and  $\theta = (\mu, \sigma, \lambda, \mu_J, \sigma_J)$ . In practice the infinite sum is truncated at  $K = 20$  terms. Maximisation of equation (3.7) is performed numerically via L-BFGS-B, with initial estimates derived from the GBM calibration.

### 3.2.5 Simulation Algorithm

For each time step  $[t_i, t_{i+1}]$  with  $\Delta t = t_{i+1} - t_i$ :

1. Draw  $N_i \sim \text{Poisson}(\lambda \Delta t)$ .
2. Draw  $Z_i \sim \mathcal{N}(0, 1)$  (diffusion shock).
3. The jump contribution is  $\sum_{k=1}^{N_i} \ln Y_k \sim \mathcal{N}(N_i \mu_J, N_i \sigma_J^2)$ .
4. The log-price increment is  $\Delta L_i = (\mu - \lambda \kappa - \frac{1}{2} \sigma^2) \Delta t + \sigma \sqrt{\Delta t} Z_i + N_i \mu_J + \sqrt{N_i} \sigma_J Z_i^J$ , where  $Z_i^J \sim \mathcal{N}(0, 1)$ , independent of  $Z_i$ .

### 3.3 Heston Stochastic Volatility Model

#### 3.3.1 Motivation

As established in Section 2.3, Bitcoin returns exhibit significant volatility clustering. The Heston (1993) model [6] addresses this by treating the instantaneous variance  $v_t$  as a separate stochastic process with mean-reversion.

**Assumption 3.7** (Heston Model Assumptions). (H1) The log-price increments are conditionally Gaussian with time-varying variance:  $d \ln S_t \mid v_t \sim \mathcal{N}((\mu - \frac{1}{2} v_t) dt, v_t dt)$ .

(H2) The instantaneous variance  $v_t \geq 0$  is a mean-reverting stochastic process driven by a second Brownian motion  $W_t^{(2)}$ .

(H3) The two Brownian motions are correlated:  $d \langle W^{(1)}, W^{(2)} \rangle_t = \rho dt$  for a constant  $\rho \in (-1, 1)$ .

(H4) The variance process is driven by a Cox-Ingersoll-Ross (CIR) SDE with mean-reversion level  $\theta > 0$ , speed  $\kappa > 0$ , and vol-of-vol  $\xi > 0$ .

#### 3.3.2 Model Formulation

**Proposition 3.8** (Heston SDE System). *Under Assumption 3.7, the price-variance system satisfies*

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \quad (3.8)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \quad (3.9)$$

where  $\text{Corr}(W^{(1)}, W^{(2)}) = \rho$ . Equation (3.9) is the Cox-Ingersoll-Ross (CIR) process [7].

#### 3.3.3 Properties of the CIR Variance Process

**Proposition 3.9** (Feller Condition). *The CIR process (3.9) satisfies  $v_t > 0$  almost surely for all  $t > 0$  if and only if the Feller condition holds:*

$$2\kappa\theta > \xi^2.$$

*When the Feller condition is violated,  $v_t$  can reach zero, though it is immediately reflected back due to the  $\sqrt{v_t}$  diffusion coefficient.*

*Remark 3.10.* The Feller condition is an important constraint on the Heston parameters. In our calibration procedure, we enforce it as a soft constraint by capping  $\xi$  at  $0.9\sqrt{2\kappa\theta}$ .

**Proposition 3.11** (Conditional Moments of the CIR Process). *For the CIR process with  $v_0 > 0$ , the conditional mean and variance at time  $t$  are:*

$$\mathbb{E}[v_t | v_0] = \theta + (v_0 - \theta)e^{-\kappa t}, \quad (3.10)$$

$$\text{Var}[v_t | v_0] = v_0 \frac{\xi^2}{\kappa} e^{-\kappa t} (1 - e^{-\kappa t}) + \frac{\theta \xi^2}{2\kappa} (1 - e^{-\kappa t})^2. \quad (3.11)$$

As  $t \rightarrow \infty$ ,  $v_t$  converges in distribution to a Gamma distribution with mean  $\theta$  and variance  $\theta \xi^2 / (2\kappa)$ .

The mean-reversion structure of the CIR process directly captures the empirical observation that Bitcoin volatility, while time-varying, tends to revert to a long-run level  $\theta$  at speed  $\kappa$ .

### 3.3.4 Parameter Calibration via Moment Matching

The Heston model has five parameters  $(\mu, \kappa, \theta, \xi, \rho)$  plus the initial variance  $v_0$ . We use a moment-matching procedure based on the realised variance (RV) series.

1. **Drift:**  $\hat{\mu} = \bar{r}/\Delta t + \hat{\sigma}^2/(2\Delta t)$ , where  $\hat{\sigma}^2 = \widehat{\text{Var}}[r_t]$ .
2. **Initial and long-run variance:**  $\hat{v}_0 = \hat{\sigma}^2/\Delta t$ ;  $\hat{\theta} = \overline{RV}$ , the sample mean of the realised variance series  $RV_t = n^{-1} \sum_{i \in \text{window}} r_i^2/\Delta t$ .
3. **Mean-reversion speed:** Fit an AR(1) model  $RV_{t+1} = \alpha_0 + \beta_1 RV_t + \varepsilon_t$ . Then  $\hat{\kappa} = -\ln(\hat{\beta}_1)/\Delta t$ .
4. **Vol-of-vol:**  $\hat{\xi} = \widehat{\text{Std}}[\Delta RV_t]/(\sqrt{\hat{\theta}} \sqrt{\Delta t})$ .
5. **Correlation:**  $\hat{\rho} = \widehat{\text{Corr}}[r_t, \Delta RV_t]$ .

### 3.3.5 Simulation via Euler-Maruyama

Because the Heston SDE does not admit a closed-form simulation scheme for arbitrary parameters, we use the Euler-Maruyama discretisation with the *full truncation* scheme to ensure non-negativity of the variance:

$$S_{t+\Delta t} = S_t \exp\left((\mu - \frac{1}{2}v_t^+)\Delta t + \sqrt{v_t^+ \Delta t} Z_1\right), \quad (3.12)$$

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t^+)\Delta t + \xi \sqrt{v_t^+ \Delta t} Z_2, \quad (3.13)$$

where  $v_t^+ = \max(v_t, 0)$  and  $(Z_1, Z_2)$  are bivariate standard normal with correlation  $\rho$ .

*Remark 3.12.* More accurate discretisation schemes exist, for example the Quadratic Exponential scheme of Andersen [8], which converges faster and better handles the boundary at  $v = 0$ . For the purposes of this thesis, the full truncation Euler scheme is sufficient to demonstrate the model's qualitative behaviour.

# Chapter 4

## Machine Learning Models for Price Prediction

### 4.1 Supervised Learning Framework

The ML prediction problem is a supervised regression task: given feature vector  $\mathbf{x}_t \in \mathbb{R}^p$ , predict  $y_{t+h} = S_{t+h}$ . All models minimise an empirical squared-error loss.

### 4.2 Elastic Net Regression

Elastic Net [9] solves

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha \rho \|\boldsymbol{\beta}\|_1 + \frac{\alpha(1-\rho)}{2} \|\boldsymbol{\beta}\|_2^2 \right\}. \quad (4.1)$$

It serves as the *linear baseline* and its coefficient vector  $\hat{\boldsymbol{\beta}}$  directly ranks feature importance by  $|\hat{\beta}_j|$ . Hyperparameters:  $\alpha = 1.0$ ,  $\rho = 0.5$ .

### 4.3 Random Forest

Random Forest [10] averages  $B = 200$  regression trees trained on bootstrap samples with random feature subsets. Node-impurity-decrease averaged across all trees provides a natural measure of feature importance. Hyperparameters:  $B = 200$ ,  $\text{max\_depth} = \text{None}$ .

### 4.4 Gradient Boosting

Gradient Boosting [11] builds an additive model by iteratively fitting shallow trees to the negative gradient of the squared-error loss. The shrinkage parameter  $\eta$  acts as regularisation. Hyperparameters:  $M = 200$ ,  $\eta = 0.05$ ,  $\text{max\_depth} = 3$ .

## 4.5 Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks [12] are recurrent networks designed for sequential data. At each time step  $t$  the cell maintains a state  $\mathbf{c}_t$  updated by four gating mechanisms:

$$\mathbf{f}_t = \sigma_g(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \quad (\text{forget gate}) \quad (4.2)$$

$$\mathbf{i}_t = \sigma_g(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \quad (\text{input gate}) \quad (4.3)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c), \quad (\text{cell candidate}) \quad (4.4)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (\text{cell update}) \quad (4.5)$$

$$\mathbf{o}_t = \sigma_g(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \quad (\text{output gate}) \quad (4.6)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t). \quad (\text{hidden state}) \quad (4.7)$$

**Sequence construction.** Tabular features of shape  $(n, p)$  are converted to 3-D LSTM input  $(n - L, L, p)$  using a sliding window of length  $L = 20$  bars. The  $i$ -th sample spans bars  $[i, i + L)$  and its label is the target at bar  $i + L$ .

**Architecture and training.** The network consists of a single LSTM layer with 64 hidden units, a Dropout layer (rate 0.2), and a Dense(1) output. Features and targets are normalised to  $[0, 1]$  via Min-Max scaling before training. Training uses the Adam optimiser [13] (learning rate  $10^{-3}$ ) with MSE loss and early stopping (patience = 5, monitor validation loss).

Table 4.1 summarises the hyperparameters of all models.

Table 4.1 – Hyperparameter settings for all ML models

| Model                        | Hyperparameter                   | Value              |
|------------------------------|----------------------------------|--------------------|
| Elastic Net                  | Regularisation strength $\alpha$ | 1.0                |
|                              | $\ell_1$ mixing ratio $\rho$     | 0.5                |
| Random Forest                | Number of trees $B$              | 200                |
|                              | Max tree depth                   | None (fully grown) |
| Gradient Boosting            | Number of trees $M$              | 200                |
|                              | Learning rate $\eta$             | 0.05               |
|                              | Max tree depth                   | 3                  |
| LSTM                         | Sequence length $L$              | 20                 |
|                              | Hidden units                     | 64                 |
|                              | Dropout rate                     | 0.2                |
|                              | Learning rate                    | $10^{-3}$          |
|                              | Max epochs (early stopping)      | 30                 |
| SDE-ML Hybrid (GB component) | Number of trees $M$              | 200                |
|                              | Learning rate $\eta$             | 0.05               |
|                              | Rolling RV window                | 20                 |

## 4.6 SDE-ML Hybrid Model with Time-Varying Drift

### 4.6.1 Motivation

The comparative analysis in Section 2.3 shows that Bitcoin log-returns exhibit structural breaks and time-varying drift, contradicting the constant-coefficient assumption of all three SDE models. Meanwhile, pure ML models achieve high point-prediction accuracy but provide no distributional forecasts and have no mechanism to quantify uncertainty.

We propose a **Hybrid SDE-ML model** that addresses both limitations simultaneously.

### 4.6.2 Model Formulation

Let  $r_t = \ln(S_t/S_{t-1})$  be the observed log-return at time  $t$ . Define:

- $\hat{\mu}_t = f_{\text{GB}}(\mathbf{x}_t)$ : the one-step drift predicted by a Gradient Boosting regressor from feature vector  $\mathbf{x}_t$ ;
- $\hat{\sigma}_t$ : the rolling realised volatility estimated over a 20-bar window,  $\hat{\sigma}_t = \frac{1}{\sqrt{\Delta t}} \sqrt{\frac{1}{w-1} \sum_{k=0}^{w-1} r_{t-k}^2}$ .

**Definition 4.1** (Hybrid SDE-ML Dynamics). The  $h$ -step-ahead conditional distribution of the price is modelled as

$$S_{t+h} = S_t \cdot \exp\left(\hat{\mu}_t \cdot h\Delta t + \hat{\sigma}_t \cdot \sqrt{h\Delta t} \cdot Z\right), \quad Z \sim \mathcal{N}(0, 1), \quad (4.8)$$

where  $\hat{\mu}_t$  is the ML-predicted drift and  $\hat{\sigma}_t$  is the realised-volatility estimate at time  $t$ .

*Remark 4.2.* The point forecast (conditional median,  $Z = 0$ ) is  $\hat{S}_{t+h} = S_t \cdot \exp(\hat{\mu}_t \cdot h\Delta t)$ . The conditional distribution is log-normal with time-varying parameters  $(\hat{\mu}_t, \hat{\sigma}_t)$ , enabling proper uncertainty quantification.

### 4.6.3 Feature Vector for the Drift Predictor

The ML drift predictor  $f_{\text{GB}}$  is trained to predict the next-bar log-return from the following features:

- lagged log-returns  $r_{t-1}, \dots, r_{t-10}$ ;
- current realised volatility  $\hat{\sigma}_t$ ;
- 14-bar RSI momentum proxy: fraction of positive returns in  $[t-14, t)$ .

### 4.6.4 Relationship to Neural SDEs

The Hybrid SDE-ML model is a specific instance of the Neural SDE family [14], in which the drift function  $b(t, S_t)$  is replaced by a non-parametric machine learning approximator trained on data. Unlike full Neural SDEs, which parameterise both drift and diffusion as neural networks, our approach uses a simpler gradient boosting regressor for the drift and an analytic realised variance estimator for



the diffusion. This provides interpretable feature importances for the drift component (Table 6.3) while retaining the theoretical guarantees of GBM for the diffusion term.

# Chapter 5

## Methodology and Data

### 5.1 Data Description

#### 5.1.1 Source and Collection

Bitcoin price data are obtained from Yahoo Finance via the `yfinance` Python library. The data include High, Low, Close, and Volume across six time resolutions: 1m, 5m, 15m, 30m, 60m, and 1d. Due to Yahoo Finance retention limits, intraday data at 15-minute resolution covers 60 calendar days. Daily data spans 2014-04-16 to 2026-04-14 (4 365 daily observations after preprocessing). The primary evaluation data set is the 15-minute series, yielding 5 664 bars.

#### 5.1.2 Data Preprocessing

1. Multi-index columns from `yfinance` are flattened to `BTC_Close`, `BTC_High`, `BTC_Low`, `BTC_Volume`.
2. Backward-fill applied to weekend gaps in daily data; remaining missing values dropped.
3. Log-returns  $r_t = \ln(S_t/S_{t-1})$  computed for SDE calibration.

#### 5.1.3 Descriptive Statistics

Table 5.1 summarises the daily Bitcoin log-return series.

Table 5.1 – Descriptive statistics: daily Bitcoin log-returns ( $r_t$ )

| Statistic                   | Value                        |
|-----------------------------|------------------------------|
| Observations                | 4 365                        |
| Mean (annualised)           | +127 %                       |
| Std. Dev. (annualised)      | 72 %                         |
| Minimum                     | −46.5 %                      |
| Maximum                     | +25.2 %                      |
| Skewness                    | −0.31                        |
| Excess kurtosis             | 7.84                         |
| Jarque-Bera $p$ -value      | < 0.001                      |
| ADF $p$ -value (log-price)  | 0.43 (fail to reject $H_0$ ) |
| ADF $p$ -value (log-return) | < 0.001 (reject $H_0$ )      |

## 5.2 Feature Engineering

For all ML models and the Hybrid, the following feature set is constructed on the 15-minute data for the  $h = 10$ -step-ahead horizon.

The choice  $h = 10$  corresponds to a 150-minute forecast horizon. This horizon is long enough to avoid a trivial one-bar-ahead nowcast, but still short enough to remain relevant for intraday Bitcoin dynamics and for the available 15-minute feature set. It therefore provides a practical middle ground between very short-horizon microstructure noise and much longer horizons where the current intraday covariates become less informative.

- **Price lags:**  $S_t, S_{t-1}, \dots, S_{t-9}$  (10 lags).
- **Volume lags:**  $V_t, \dots, V_{t-4}$  (5 lags).
- **High and Low:**  $H_t, L_t$ .
- **Technical indicators:**
  - $\text{SMA}_{20} = \frac{1}{20} \sum_{i=0}^{19} S_{t-i}$ ;
  - $\text{RSI}_{14}$ : 14-period Relative Strength Index;
  - High-Low range:  $HL_t = H_t - L_t$ .

The LSTM and Hybrid models use the same features; the Hybrid's drift predictor additionally uses lagged log-returns and the rolling realised volatility as described in Section 4.6.

This asymmetry should be interpreted carefully. The SDE models are structural benchmarks and are intentionally calibrated from the price process alone, without additional explanatory variables such as volume or technical signals. The ML models, by contrast, are designed to exploit a richer information set. Therefore, the purpose of the comparison is not to claim that all model classes operate under identical informational conditions, but to analyse what each paradigm can deliver under its natural assumptions. In this sense, the hybrid models provide the most informative bridge: they retain an SDE-based dynamic structure while using a feature set comparable to the one used by the ML baselines.

## 5.3 Walk-Forward Cross-Validation

A simple chronological 80/20 split is susceptible to look-ahead bias and provides only a single estimate of out-of-sample performance. We therefore adopt **expanding-window walk-forward cross-validation** with  $K = 5$  folds.

**Protocol.** Let  $n$  be the total number of observations. The initial training set comprises the first  $\lfloor 0.5n \rfloor$  observations. In fold  $k$  ( $k = 1, \dots, K$ ), the training set expands by one block of size  $\lfloor 0.5n/K \rfloor$  and the following block is used as the test set. Table 5.2 shows the fold boundaries for the 15-minute series.

Table 5.2 – Walk-forward fold boundaries (15-minute data,  $n = 5,664$ )

| Fold | Train start | Train end | Test bars |
|------|-------------|-----------|-----------|
| 1    | 0           | 2 832     | 566       |
| 2    | 0           | 3 398     | 566       |
| 3    | 0           | 3 964     | 566       |
| 4    | 0           | 4 530     | 566       |
| 5    | 0           | 5 096     | 568       |

This design ensures that every prediction is made strictly out-of-sample and that later folds test on more recent, potentially more volatile data. Fold 5 contains 568 test bars rather than 566 because the remainder from partitioning the second half of the sample is allocated to the final fold.

## 5.4 Evaluation Metrics

### 5.4.1 Regression Metrics

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (5.1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (5.2)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}. \quad (5.3)$$

### 5.4.2 Distributional Metrics (SDE and Hybrid models)

- KS statistic: Kolmogorov-Smirnov distance between simulated and empirical return distributions.
- Coverage: empirical coverage probability of the simulated 90 % prediction interval.

### 5.4.3 Economic Metrics

To assess practical trading value, we implement a **simple momentum strategy**: at each bar  $t$ , take a long position if the model predicts an increase ( $\hat{S}_{t+h} > S_t$ ), otherwise remain flat. Transaction costs of 0.1 % per trade are applied.

Let  $r_t^{\text{strat}}$  denote the per-bar strategy return. We report:

1. **Total return**:  $(C_T - C_0)/C_0$ , where  $C_t$  is portfolio value at bar  $t$ .
2. **Annualised Sharpe ratio**:

$$\text{SR} = \frac{\bar{r}^{\text{strat}}}{\hat{\sigma}^{\text{strat}}} \cdot \sqrt{365 \times 96}$$

(Bitcoin trades 24/7; 96 fifteen-minute bars per day).

3. **Maximum drawdown**:  $\text{MDD} = \min_t \frac{C_t - \max_{s \leq t} C_s}{\max_{s \leq t} C_s}$ .

# Chapter 6

## Experimental Results

### 6.1 SDE Model Calibration Results

#### 6.1.1 GBM Calibration

On the daily return series ( $\Delta t = 1$  day):

$$\hat{\mu} = 0.00186 \text{ day}^{-1} \quad (= 67.9\% \text{ ann.}), \quad \hat{\sigma} = 0.0427 \text{ day}^{-1/2} \quad (= 68.0\% \text{ ann.}).$$

KS test:  $D = 0.072$ ,  $p = 0.003$  (significant misfit due to fat tails).

#### 6.1.2 Merton Jump-Diffusion Calibration

MLE yields:  $\hat{\mu} = 0.00312$ ,  $\hat{\sigma} = 0.0388$ ,  $\hat{\lambda} = 0.84$ ,  $\hat{\mu}_J = -0.0087$ ,  $\hat{\sigma}_J = 0.0231$ . KS:  $D = 0.041$ ,  $p = 0.11$  (fail to reject at 5 %).

#### 6.1.3 Heston Model Calibration

In the final implementation, the Heston model is calibrated directly on the 15-minute series used for forecasting, with  $\Delta t = 1/96$  day, rather than being estimated on daily data and then mechanically rescaled to intraday frequency. The moment-matching procedure is applied to rolling realised variance computed at the same sampling frequency. This makes the comparison with the 15-minute ML and Hybrid forecasts methodologically cleaner.

Using the 30-bar rolling realised variance series, the calibration yields  $\hat{\kappa} = 0.231$ ,  $\hat{\theta} = 0.00176$ ,  $\hat{\xi} = 0.0124$ ,  $\hat{\rho} = -0.132$ . In all SDE forecast experiments, 1 000 Monte Carlo paths are simulated per forecast origin, and point predictions are taken from the median simulated path. The implied variance half-life is  $\ln 2 / \hat{\kappa} \approx 3$  trading days.

## 6.2 Machine Learning Model Results

### 6.2.1 Single Train-Test Split

Table 6.1 reports performance on the chronological 80/20 held-out test set (15-minute data,  $h = 10$  steps ahead).

Table 6.1 – Test set performance: single split (15-minute data,  $h = 10$ )

| Model                   | MAE (USD)    | RMSE (USD)   | $R^2$        | Sharpe      |
|-------------------------|--------------|--------------|--------------|-------------|
| Elastic Net             | 337.6        | 491.5        | 0.963        | 1.47        |
| Random Forest           | 392.9        | 542.5        | 0.954        | 1.38        |
| Gradient Boosting       | 351.5        | 502.0        | 0.961        | 1.52        |
| LSTM                    | 318.4        | 463.7        | 0.966        | 1.61        |
| <b>SDE-ML Hybrid</b>    | <b>298.1</b> | <b>441.2</b> | <b>0.968</b> | <b>1.72</b> |
| GBM (SDE)               | 1 214.3      | 1 687.2      | 0.704        | 0.89        |
| Merton Jump-Diff. (SDE) | 1 189.7      | 1 653.8      | 0.717        | 0.94        |
| Heston (SDE)            | 1 201.5      | 1 670.1      | 0.710        | 0.91        |
| Buy & Hold              | —            | —            | —            | 1.21        |

For interpretation, a naive persistence benchmark is also computed in the experimental notebook: the forecast simply sets  $\hat{S}_{t+h} = S_t$ . This is an important reference in a near-random-walk market, because many apparently strong models mainly reproduce the last observed price. The feature-importance results in Section 6.3 should therefore be read relative to this baseline rather than in isolation.

Paired  $t$ -tests on absolute forecast errors were also conducted in the notebook to assess whether observed differences between the main ML and Hybrid specifications are statistically significant. Because the exact  $p$ -values are reported in the computational notebook rather than reproduced here, the thesis uses these tests as a robustness check and avoids stronger claims than the available evidence supports.

### 6.2.2 Walk-Forward Cross-Validation Results

Table 6.2 reports fold-level MAE and the cross-validation mean  $\pm$  standard deviation across the five folds. Compared to the single split, all models show modestly higher MAE (as expected for a more rigorous out-of-sample protocol), and the relative ranking is preserved.

Table 6.2 – Walk-forward cross-validation: MAE per fold (USD, 15-minute data)

| Model                | Fold 1     | Fold 2     | Fold 3     | Fold 4     | Fold 5     | Mean       | Std       |
|----------------------|------------|------------|------------|------------|------------|------------|-----------|
| Elastic Net          | 389        | 421        | 398        | 412        | 368        | 398        | 19        |
| Random Forest        | 447        | 489        | 461        | 478        | 432        | 461        | 21        |
| Gradient Boosting    | 412        | 438        | 419        | 431        | 396        | 419        | 16        |
| LSTM                 | 371        | 408        | 381        | 402        | 356        | 384        | 20        |
| <b>SDE-ML Hybrid</b> | <b>342</b> | <b>378</b> | <b>359</b> | <b>371</b> | <b>328</b> | <b>356</b> | <b>19</b> |

The low standard deviation of fold-level MAE (16–21 USD) indicates stable out-of-sample performance. The Hybrid SDE-ML model achieves the best mean MAE (356 USD) with comparable vari-

ability in this table, but the broader interpretation should remain cautious: on short-horizon Bitcoin forecasting, small differences between strong lag-based models may not always translate into statistically decisive improvements once paired error tests are applied.

## 6.3 Feature Importance Analysis

Table 6.3 reports the top-10 feature importances from the Gradient Boosting model, which achieves the best interpretable tree-based performance. Importances are measured by the average decrease in MSE across all trees and splits.

Table 6.3 – Gradient Boosting: top-10 feature importances

| Feature            | Importance | Cumulative |
|--------------------|------------|------------|
| BTC_Close (lag 1)  | 0.342      | 0.342      |
| BTC_Close (lag 2)  | 0.187      | 0.529      |
| BTC_Close (lag 3)  | 0.089      | 0.618      |
| BTC_Close (lag 4)  | 0.064      | 0.682      |
| BTC_High           | 0.058      | 0.740      |
| BTC_Low            | 0.051      | 0.791      |
| SMA <sub>20</sub>  | 0.047      | 0.838      |
| BTC_Close (lag 5)  | 0.038      | 0.876      |
| BTC_Volume (lag 1) | 0.032      | 0.908      |
| RSI <sub>14</sub>  | 0.028      | 0.936      |

The dominance of the most recent close price (lag 1, importance 0.342) is consistent with the near-random-walk character of Bitcoin prices: the best linear predictor of  $S_{t+10}$  is largely determined by  $S_t$ . However, the combined weight of lags 2–5 (0.378) and technical indicators (SMA, RSI, 0.075) demonstrates that past price history and momentum signals carry incremental predictive information beyond a simple last-value baseline.

## 6.4 Economic Evaluation

Table 6.4 presents the economic performance of the momentum strategy across all models, benchmarked against buy-and-hold.

Table 6.4 – Economic performance: momentum strategy (15-minute test set)

| Strategy             | Total Return (%) | Sharpe      | Max DD (%)  | PnL (USD)   |
|----------------------|------------------|-------------|-------------|-------------|
| Buy & Hold           | +4.2             | 1.21        | −8.7        | +420        |
| Elastic Net          | +3.1             | 1.34        | −5.2        | +310        |
| Random Forest        | +2.8             | 1.28        | −5.9        | +280        |
| Gradient Boosting    | +3.4             | 1.47        | −4.8        | +340        |
| LSTM                 | +3.6             | 1.58        | −4.1        | +360        |
| <b>SDE-ML Hybrid</b> | <b>+3.9</b>      | <b>1.72</b> | <b>−3.7</b> | <b>+390</b> |

**Discussion.** All ML strategies show a lower absolute return than buy-and-hold (4.2 % vs. 3.1 – 3.9 %), reflecting the cost of being flat during positive price moves that were not predicted. However, all ML

strategies achieve *higher Sharpe ratios* and substantially smaller maximum drawdowns. The buy-and-hold strategy suffers a maximum drawdown of 8.7 %, while the Hybrid SDE-ML model limits drawdown to 3.7 %.

The Hybrid SDE-ML model achieves the best risk-adjusted performance ( $\text{Sharpe} = 1.72$ ), a 42 % improvement over buy-and-hold ( $\text{Sharpe} = 1.21$ ). This demonstrates that the accuracy gains of the hybrid model translate into economically meaningful improvements in portfolio risk management within this stylised backtest. The result should not be read as a claim of deployable trading profitability, and it should be considered together with the naive baseline and the statistical-significance checks discussed above.

## 6.5 Distributional Forecast Quality (SDE and Hybrid Models)

Coverage probabilities of the simulated 90 % prediction interval:

- GBM: 74.2 % (significant undercoverage due to fat tails);
- Merton Jump-Diffusion: 91.3 % (closest to nominal);
- Heston: 88.7 % (slight undercoverage);
- **SDE-ML Hybrid: 91.7 % (best overall coverage).**

The Hybrid model achieves the strongest distributional coverage among the models reported here. This supports the main methodological argument of the thesis: even when a pure ML baseline is highly competitive on point accuracy, the hybrid framework remains valuable because it combines data-driven adaptivity with simulation-based distributional forecasts.

## 6.6 Comparative Analysis

### 6.6.1 Predictive Accuracy

The empirical comparison should be read at two levels. First, classical SDE specifications remain useful structural benchmarks, but they are less accurate than feature-based ML models on short-horizon point forecasting. Second, the best-performing ML and Hybrid variants are relatively close to each other, so the main conclusion is not that one architecture universally dominates, but that the hybrid approach is competitive while preserving distributional interpretability that pure point-forecast models do not provide.

### 6.6.2 Computation and Scalability

Elastic Net and Gradient Boosting are the fastest to train ( $< 10$  s for all 5 000 training samples). Random Forest is slightly slower due to bootstrapping. The LSTM requires approximately 45 seconds per fold on CPU; the Hybrid model adds negligible overhead over Gradient Boosting.



### **6.6.3 Interpretability**

SDE models offer the most direct economic interpretation (drift, vol, jump frequency). Among ML models, Elastic Net coefficients and tree-based feature importances (Table 6.3) provide interpretable rankings. The Hybrid model inherits the interpretability of Gradient Boosting for the drift component and the theoretical structure of GBM for the noise component.

# Conclusion

## Summary of Results

This thesis addressed Bitcoin price prediction from the complementary perspectives of mathematical finance (SDE models) and data science (ML models), with the following principal findings.

1. **Theoretical justification.** Via a series of propositions, we established that Bitcoin prices admit a stochastic process representation under the Efficient Market Hypothesis, and that the observed stylised facts (heavy tails, volatility clustering, structural breaks) motivate progressive extensions beyond GBM: the Merton Jump-Diffusion model and the Heston Stochastic Volatility model.
2. **SDE calibration.** All three SDE models were successfully calibrated, with the final Heston implementation estimated directly at the 15-minute frequency used in forecasting rather than via daily-to-intraday rescaling. The Merton model best fits the empirical return distribution (KS  $p = 0.11$ ) and achieves closest-to-nominal prediction interval coverage (91.3 %).
3. **ML models remain strong short-horizon baselines.** On the 15-minute forecasting task, simple lag-based machine learning models are difficult to beat. This is an important empirical result, because it shows that short-horizon Bitcoin prediction is highly sensitive to feature timing, horizon definition, and evaluation protocol.
4. **Walk-forward validation.** The walk-forward results confirm that the relative model ranking is robust across different time periods (standard deviations of fold-level MAE: 16–21 USD), and are approximately 15–20 % higher in absolute terms than single-split estimates, as expected for a more conservative evaluation protocol.
5. **Hybrid SDE+ML models provide a meaningful methodological bridge.** The implemented hybrid family combines a stochastic-process view of price dynamics with a data-driven approximation of the drift component. Even when a hybrid model is not the best model under every point metric, it remains valuable because it preserves distributional interpretation while improving adaptivity relative to classical constant-coefficient SDEs.
6. **Economic evaluation is informative but simplified.** The trading comparison shows that predictive differences can translate into different Sharpe ratios and drawdowns, but these results should be treated as comparative indicators rather than as direct evidence of deployable profitability.

7. **Feature importance.** The most recent close price (lag 1) is the dominant predictor (importance 0.342), consistent with the near-random-walk character of Bitcoin. Technical indicators (SMA, RSI) and volume contribute incremental predictive power.

## Discussion

The study reveals a fundamental tension between the two modelling paradigms. SDE models provide interpretable economic parameters, closed-form distributional forecasts, and theoretical grounding, but their constant-coefficient assumptions prevent them from tracking short-term momentum. ML models excel at point prediction by exploiting non-linear feature patterns, but are atheoretical and provide no distributional output.

The hybrid SDE+ML framework addresses this tension by using machine learning to parameterise selected components of an otherwise interpretable stochastic model. The main contribution of the thesis is therefore not only a single numerical result, but a reproducible framework for studying how structural stochastic models and flexible data-driven models can be combined. This framing is more accurate and more defensible than the stronger claim that one hybrid variant must dominate all baselines under every metric and every split.

## Limitations and Future Work

1. **Regime-switching SDE.** A Markov-switching SDE would better capture the structural breaks in Bitcoin's price history.
2. **Full Neural SDE.** Parameterising both drift *and* diffusion as neural networks would extend the hybrid approach and improve adaptivity.
3. **Kalman filtering and state-space estimation.** A Kalman-filter layer or a more general state-space formulation could improve latent-state estimation and provide a cleaner connection between noisy observations and the underlying stochastic dynamics.
4. **Order book data.** High-frequency limit order book features (bid-ask spread, depth imbalance) are important at sub-minute horizons.
5. **Multi-asset models.** Extending to multivariate SDEs or multi-output ML would incorporate BTC correlations with other cryptocurrencies and macro assets.
6. **Sentiment and on-chain data.** Social media sentiment, on-chain volume, and hash rate are informative signals not yet included.